

IT 462:
Big Data Systems
Course Project
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Diabetes Health Indicators

Proposal

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1. Dataset Identification

Dataset Name: CDC Diabetes Health Indicators

Domain: Healthcare & Public Health

Source URL: <https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>

File Used: diabetes_binary_health_indicators_BRFSS2015.csv

Dataset Size: 253,680 rows | 21 Features + 1 Target | ~22 MB (CSV)

Dataset Scale Note: At 253,680 records, this dataset meets and exceeds the course minimum requirement of 50,000 rows. While 500,000+ rows is noted as preferred for optimal Spark scalability, the dataset's rich 21-feature schema, mixed data types, and class imbalance characteristics make it genuinely suitable for distributed processing with Apache Spark MLlib.

2. Data Schema

The dataset contains 21 attributes covering health indicators, lifestyle factors, and socioeconomic variables. The target variable is a three-class ordinal label reflecting diabetes status.

Variable Name	Role	Type	Attribute Group	Description	Example Values
Diabetes_binary	Target	Ordinary	Target Variable	Diabetes or prediabetes diagnosis	0 = No diabetes 1 = Prediabetes 2=Diabetes
HighBP	Feature	Binary	Health Indicator	Ever told you have high blood pressure	0 = No, 1 = Yes
HighChol	Feature	Binary	Health Indicator	Ever told you have high cholesterol	0 = No, 1 = Yes
CholCheck	Feature	Binary	Health Indicator	Cholesterol check in past 5 years	0 = No, 1 = Yes
Stroke	Feature	Binary	Health Indicator	Ever told you had a stroke	0 = No, 1 = Yes
HeartDiseaseorAttack	Feature	Binary	Health Indicator	Coronary heart disease or myocardial infarction	0 = No, 1 = Yes
DiffWalk	Feature	Binary	Health Indicator	Serious difficulty walking or climbing stairs	0 = No, 1 = Yes
Smoker	Feature	Binary	Lifestyle & Personal	Smoked at least 100 cigarettes in lifetime	0 = No, 1 = Yes
PhysActivity	Feature	Binary	Lifestyle & Personal	Physical activity in past 30 days (excl. job)	0 = No, 1 = Yes

Fruits	Feature	Binary	Lifestyle & Personal	Consume fruit 1 or more times per day	0 = No, 1 = Yes
Veggies	Feature	Binary	Lifestyle & Personal	Consume vegetables 1 or more times per day	0 = No, 1 = Yes
HvyAlcoholConsump	Feature	Binary	Lifestyle & Personal	Heavy drinker (men >14 drinks/wk; women >7 drinks/wk)	0 = No, 1 = Yes
Sex	Feature	Binary	Lifestyle & Personal	Biological sex	0 = Female, 1 = Male
AnyHealthcare	Feature	Binary	Social & Economic	Has any health care coverage (insurance, HMO, etc.)	0 = No, 1 = Yes
NoDocbcCost	Feature	Binary	Social & Economic	Could not see doctor in past 12 months due to cost	0 = No, 1 = Yes
GenHlth	Feature	Integer	Ordinal Scale	Self-rated general health status	1=Excellent → 5=Poor
Age	Feature	Integer	Ordinal Scale	The dataset groups age into 13 categorical levels. The first category represents ages 18–24, followed by categories in approximately 4-year increments (25–29, 30–34, 35–39, ...) up to the final category, which represents ages 80 and above (80+)	1=18–24, 13=80+
Education	Feature	Integer	Ordinal Scale	Highest education level attained	1=No school → 6=College grad
Income	Feature	Integer	Ordinal Scale	Annual household income bracket	1=<\$10k → 8=≥\$75k
BMI	Feature	Integer	Continuous / Count	Body Mass Index	e.g., 25 (normal), 30 (obese)
MentHlth	Feature	Integer	Continuous / Count	Days mental health not good in past 30 days	Scale: 1–30 days
PhysHlth	Feature	Integer	Continuous / Count	Days physical health not good in past 30 days	Scale: 1–30 days

Table 1. Dataset Key Features

3. Problem Statement

The global healthcare system faces a critical gap in detection where millions of people remain undiagnosed with diabetes until they experience serious health complications [1]. Medical screening is often difficult to access because of high costs and the requirement for specialized facilities, which creates a major barrier to early intervention [2]. This study investigates whether data provided directly by patients regarding their daily habits, activity levels, and economic status

can accurately identify individuals at risk before their symptoms become severe. By closing this information gap, public health officials can distribute resources to vulnerable communities more effectively, and doctors can use these findings to prioritize patients who need immediate testing.

4. Motivation

The motivation for this work is the necessary shift toward prevention as chronic diseases continue to exhaust global medical resources. By using the CDC's Behavioral Risk Factor Surveillance System (BRFSS), we can use large amounts of existing survey data to find hidden patterns in how these diseases develop over time [1]. Successfully addressing this challenge proves that Big Data frameworks like Spark are essential for solving large public health issues. This approach goes beyond simple business analytics to provide insights that can encourage people to change their behavior and reduce the long term impact of chronic illness [3].

5. Suitability for Spark

This project is a strong fit for Apache Spark for the following reasons:

- **Scale:** Training a multi-class classification model on over 253,000 records with 21 mixed-type features (binary, ordinal, numerical) benefits from Spark MLlib's distributed in-memory processing. While the dataset exceeds the course's 50,000-row minimum, the pipeline is designed to scale linearly to the full multi-year CDC BRFSS repository (millions of records) without code changes.
- **Class Imbalance Handling:** The three-class label (0/1/2) exhibits significant imbalance, requiring distributed stratified sampling and reweighting, operations well-suited to Spark's partitioned data model.
- **Hyperparameter Tuning:** Spark's CrossValidator and ParamGridBuilder allow efficient distributed grid search across model configurations, which would be prohibitively slow on a single machine.
- **Pipeline Extensibility:** The Spark ML Pipeline design ensures that adding more years of BRFSS data or additional features requires only data ingestion changes, not architectural rewrites.

References

- [1] CDC, "Diabetes Health Indicators Dataset," UCI Machine Learning Repository, 2023. [Online]. Available: <https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>. [Accessed: Feb. 13, 2026].
- [2] Centers for Disease Control and Prevention (CDC), "Behavioral Risk Factor Surveillance System (BRFSS) Survey Data," 2015. [Online]. Available: https://www.cdc.gov/brfss/annual_data/annual_2015.html.
- [3] World Health Organization, "Global Report on Diabetes," Geneva, Switzerland, 2016. [Online]. Available: <https://www.who.int/publications/i/item/9789241565257>.